MODULE 5 BRAINSTEM

The Brainstem.

THE CENTRAL NERVOUS SYSTEM, THE BRAINSTEM

A reference for the brainstem video and lab. This page covers the three regions of the brainstem, the midbrain, the pons, and the medulla oblongata, the structures within each, and the reticular formation that runs through them. The focus is on the structures and the job each one does.

BY THE END

1. Name the three regions of the brainstem and locate each one.
2. Identify the key structures of the midbrain, the pons, and the medulla oblongata.
3. Describe the reticular formation and the reticular activating system.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The brainstem

drsrennie-stack.github.io/new-build-bio4-solano/brainstem-concept-videos.html

The Brainstem, an Overview

The brainstem connects the cerebrum, the diencephalon, and the cerebellum to the spinal cord. Every ascending and descending tract passes through it, and it holds the nuclei for ten of the twelve cranial nerves.

THE THREE REGIONS AND WHAT RUNS THROUGH THEM

Structure	What it is
Brainstem	The stalk of the brain connecting the higher regions to the spinal cord; the midbrain, pons, and medulla
Midbrain	The most superior region of the brainstem, also called the mesencephalon
Pons	The middle region of the brainstem, a bridge to the cerebellum
Medulla oblongata	The most inferior region of the brainstem, continuous with the spinal cord
Cranial nerve nuclei	The clusters of gray matter in the brainstem that give rise to or receive cranial nerves III through XII
Passing tracts	The brainstem carries every ascending sensory tract and descending motor tract between the brain and the cord

The Midbrain

The midbrain, or mesencephalon, is the short, most superior region of the brainstem. It surrounds the cerebral aqueduct.

THE STRUCTURES OF THE MIDBRAIN

Structure	What it is
Tectum	The roof of the midbrain, made of four rounded bumps called colliculi
Superior colliculi	The upper pair of colliculi; centers for visual reflexes, such as turning the eyes and head toward a sight
Inferior colliculi	The lower pair of colliculi; centers for auditory reflexes, such as turning toward a sound
Cerebral peduncles	The paired stalks on the front of the midbrain that carry descending motor tracts
Substantia nigra	A darkly pigmented midbrain nucleus that helps regulate movement, working with the basal ganglia
Cerebral aqueduct	The narrow channel through the midbrain that carries cerebrospinal fluid from the third to the fourth ventricle
Cranial nerve nuclei	The midbrain holds the nuclei for CN III, the oculomotor nerve, and CN IV, the trochlear nerve

HOLD ONTO THIS

Superior colliculi sit above and serve sight; inferior colliculi sit below and serve sound. Upper for eyes, lower for ears, and both are reflex centers, not the places where you become conscious of the sight or the sound.

The Pons

The pons is the rounded, middle region of the brainstem. Its name means bridge.

THE STRUCTURES OF THE PONS

Structure	What it is
Pons	The middle region of the brainstem; a bridge between the medulla, the midbrain, and the cerebellum
Pontine nuclei	Clusters of gray matter that relay motor information from the cerebral cortex to the cerebellum
Middle cerebellar peduncles	The thick stalks that carry signals from the pontine nuclei into the cerebellum
Fourth ventricle	The cerebrospinal fluid space that lies behind the pons and the upper medulla
Cranial nerve nuclei	The pons holds the nuclei for CN V (trigeminal), CN VI (abducens), CN VII (facial), and CN VIII (vestibulocochlear)

The Medulla Oblongata

The medulla oblongata is the most inferior region of the brainstem. It is continuous with the spinal cord at the foramen magnum and contains centers that keep the body alive.

THE STRUCTURES AND VITAL CENTERS OF THE MEDULLA

Structure	What it is
Medulla oblongata	The most inferior region of the brainstem; it merges into the spinal cord
Pyramids	A pair of ridges on the front of the medulla formed by the descending corticospinal motor tracts
Decussation of the pyramids	The crossover point where most corticospinal fibers pass to the opposite side; it is why one hemisphere controls the opposite side of the body
Cardiac center	The medullary center that adjusts the rate and force of the heartbeat
Respiratory center	The medullary center that sets the basic rhythm of breathing
Vasomotor center	The medullary center that adjusts the diameter of blood vessels, and so blood pressure
Cranial nerve nuclei	The medulla holds nuclei for CN VIII (cochlear), CN IX, CN X, CN XI, and CN XII

The Three Regions Compared

Each region of the brainstem has its own structures and its own set of cranial nerve nuclei. Compare them.

MIDBRAIN, PONS, AND MEDULLA SIDE BY SIDE

Region	Key structures	Cranial nerve nuclei
Midbrain	The tectum, with superior colliculi for visual reflexes and inferior colliculi for auditory reflexes; the cerebral peduncles carrying descending motor tracts; the substantia nigra	III and IV
Pons	A bridge between the medulla, midbrain, and cerebellum; contains the pontine nuclei that relay motor information to the cerebellum	V, VI, VII, and VIII
Medulla oblongata	Contains the pyramids, where the corticospinal tracts decussate; holds the vital centers for heart rate, breathing, and blood vessel diameter	VIII through XII

The Reticular Formation

The reticular formation is not one structure. It is a net of gray matter cell bodies woven through the white matter of the entire brainstem.

THE NETWORK AND WHAT IT DOES

Structure	What it is
Reticular formation	A network of gray matter neurons spread through the core of the brainstem
Reticular activating system	The RAS, the ascending part of the network; it keeps the cortex awake, alert, and attentive
Functions of the RAS	Maintaining consciousness and arousal, holding attention, and filtering out unimportant incoming stimuli so the brain is not overloaded
Descending component	The part that helps set the muscle tone of resting skeletal muscle and coordinates motor activity with autonomic reflexes
Loss of RAS activity	When the ascending system is shut down the result is sleep, and severe damage can cause coma

A SMALL REGION WITH VITAL JOBS

The **medulla** holds the **cardiac**, **respiratory**, and **vasomotor** centers. Because these centers run the heartbeat and breathing, injury to the medulla, unlike injury to a single cortical area, is immediately life-threatening.

The **decussation of the pyramids** in the medulla is the anatomical reason a stroke in the right hemisphere weakens the left side of the body. The motor fibers have already crossed by the time they reach the cord.

The **reticular activating system** explains the link between the brainstem and consciousness. A patient can lose a great deal of cortex and stay awake, but damage to the small ascending reticular network can drop them into a coma. The **substantia nigra** ties the brainstem to movement: its loss is the basis of Parkinson disease.